Exercise 16

Solution 1: Statistics is the science which deals with the collection, presentation, analysis and interpretation of numerical data.

Solution 2:

The fundamental characteristics of data (statistics) are as follows:

(i) Numerical facts alone constitute data.

(ii) Qualitative characteristics like intelligence and poverty, which cannot be measured numerically, do not form data.

(iii) Data are aggregate of facts. A single observation does not form data.

(iv) Data collected for a definite purpose may not be suited for another purpose.

(v) Data in different experiments are comparable.

Solution 3: Primary data: The data collected by the investigator himself with a definite plan in mind are known as primary data.

Secondary data: The data collected by someone other than the investigator are known as secondary data.

Primary data are highly reliable and relevant because they are collected by the investigator himself with a definite plan in mind, whereas secondary data are collected with a purpose different from that of the investigator and may not be fully relevant to the investigation.

Solution 4:

(i) Variate: Any character which is capable of taking several different values is called a variant or a variable.

(ii) Class interval: Each group into which the raw data is condensed is called class interval.

(iii) Class size: The difference between the true upper limit and the true lower limit of a class is called its class size.

(iv) Class mark of a class: The average of upper and lower limit of a class interval is called its class mark.

$$\text{i.e., class mark} = \frac{(\text{Upper limit} + \text{Lower limit})}{2}$$

(v) Class limit: Each class is bounded by two figures, which are called class limits.

(vi) True class limits: In the exclusive form, the upper and lower limits of a class are respectively known as true upper limit and true lower limit.

In the inclusive form of frequency distribution, the true lower limit of a class is obtained by subtracting 0.5 from the lower limit and the true upper limit of the class is obtained by adding 0.5 to the upper limit.

(vii) Frequency of a class: Frequency of a class is the number of times an observation occurs in that class.

(viii) Cumulative frequency of a class: Cumulative frequency of a class is the sum total of all the frequencies up to and including that class.

Solution 5:

(i)

Blood group	tally marks	Number of students
A	𝍸 ||||	9
B	𝍸 |	6
O	𝍸 𝍸 ||	12
AB	|||	3

(ii) AB is rarest and O is most common.

Solution 6:

Number of heads	tally marks	Frequency
0	卌 |	6
1	卌 卌	10
2	卌 ||||	9
3	卌	5

Solution 7: The minimum observation is 0 and the maximum observation is 8.

Therefore, classes of the same size covering the given data are 0-2, 2-4, 4-6 and 6-8.

Frequency distribution table:

Class	Tally mark	Frequency
0-2	卌 卌 |	11
2-4	卌 卌 卌 ||	17
4-6	卌 ||||	9
6-8	|||	3

Solution 8:

(i)

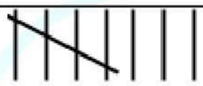
Class interval	tally marks	Frequency
0-5	卌 卌	10
5-10	卌 卌 |||	13
10-15	卌	5
15-20	||	2

(ii) As we can see from the table, there are 2 children who watched tv for 15 hours or more.

Solution 9: The minimum observation is 0 and the maximum observation is 25.

Therefore, classes of the same size covering the given data are 0-5, 5-10, 10-15, 15-20 and 20-25.

Frequency distribution table:

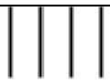
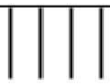
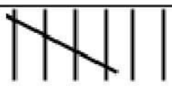
Class	Tally mark	Frequency
0-5		6
5-10		10
10-15		8
15-20		8
20-25		8

Solution 10:

The minimum observation is 6 and the maximum observation is 24.

Therefore, classes of the same size covering the given data are 6-9, 9-12, 12-15, 15-18, 18-21 and 21-24.

Frequency distribution table:

Class	Tally mark	Frequency
6-9		5
9-12		4
12-15		4
15-18		7

18-21	$\vert\vert\vert$	3
21-24	$\cancel{\vert\vert\vert\vert}\ \vert\vert$	7

Solution 11:

The minimum observation is 210 and the maximum observation is 330.

Therefore, classes of the same size covering the given data are 210-230, 230-250,250-270,270-290,290-310 and 310-330.

Frequency distribution table:

Class	Tally mark	Frequency
210-230	$\vert\vert\vert\vert$	4
230-250	$\vert\vert\vert\vert$	4
250-270	$\cancel{\vert\vert\vert\vert}$	5
270-290	$\vert\vert\vert$	3
290-310	$\cancel{\vert\vert\vert\vert}\ \vert\vert$	7
310-330	$\cancel{\vert\vert\vert\vert}$	5

Solution 12: The minimum observation is 30 and the maximum observation is 120.

Frequency distribution table:

Class	Tally mark	Frequency
30-40	$\vert\vert\vert\vert$	4

40-50	卌 I	6
50-60	III	3
60-70	卌	5
70-80	卌 IIII	9
80-90	卌 I	6
90-100	II	2
100-110	III	3
110-120	II	2

Cumulative frequency table:

Class	Tally mark	Frequency	Cumulative frequency
30-40	IIII	4	4
40-50	卌 I	6	10
50-60	III	3	13
60-70	卌	5	18
70-80	卌 IIII	9	27
80-90	卌 I	6	33
90-100	II	2	35

100-110	|||	3	38
110-120	||	2	40

Solution 13:

Class	tally marks	Frequency	Cumulative frequency
145-150	||||	4	4
150-155	~~||||~~ ||||	9	$4 + 9 = 13$
155-160	~~||||~~ ~~||||~~ ||	12	$13 + 12 = 25$
160-165	~~||||~~	5	$25 + 5 = 30$

Solution 14:

The cumulative frequency table can be presented as given below:

Age (in years)	No. of patients	Cumulative frequency
10-20	90	90
20-30	50	140
30-40	60	200
40-50	80	280
50-60	50	330
60-70	30	360

Solution 15: The grouped frequency table can be presented as given below:

Marks	No. of students
0-10	5
10-20	7
20-30	20

30-40	8
40-50	5
50-60	3

Solution 16:

The frequency table can be presented as given below:

Marks	Number of students
0-10	17
10-20	5
20-30	7
30-40	8
40-50	13
50-60	10

Solution 17:

The frequency table can be presented as below:

Class	Frequency
0-10	8
10-20	5
20-30	12
30-40	35
40-50	24
50-60	16

Solution 18:

Range = Maximum value – minimum value

$= 100 - 46$

$= 54$

Thus, the range is 54.

Solution 19:

(i) class mark of the class 90 – 120 $= \dfrac{\text{upper limit} + \text{lower limit}}{2} = \dfrac{120+90}{2} = \dfrac{210}{2} = 105$

(ii) mid-value = 10

width = 6

Let the lower limit of the class be *x*.

The, upper limit of the class = x + 6

Now,

class mark = midpoint

$\Rightarrow \dfrac{\text{upper limit} + \text{lower limit}}{2} = 10$

$\Rightarrow \dfrac{(x+6)+x}{2} = 10$

$\Rightarrow \dfrac{2x+6}{2} = 10$

$\Rightarrow 2x + 6 = 20$

$\Rightarrow 2x = 14$

$\Rightarrow x = 7$

(iii) width = 5

lower class limit of lowest class i.e., 1st class = 10

Total number of classes = 5

∴ The classes will be 10-15, 15-20, 20-25, 25-30, 30-35.

Thus, the upper class limit of the highest class = 35.

(iv) Class marks = 15, 20, 25, ...

class size = 20 − 15 = 5

Let lower limit of class be *x*.

Then the upper limit of the class = x + 5

$\therefore \frac{(x+5)+x}{2} = 20$

$\Rightarrow 2x + 5 = 40$

$\Rightarrow 2x = 35$

$\Rightarrow x = 17.5$

Now, upper class $x + 5 = 17.5 + 5 = 22$

Thus, the required class is 17.5-22.5.

(v) 20 will be included in the class interval 20-30.

Solution 20: The complete table will be

Height (in cm)	Frequency	Cumulative frequency
160 – 165	15	a = 15
165 – 170	b = 35 – 15 = 20	35
170 – 175	12	c = 35 + 12 = 47
175 – 180	d = 50 – 47 = 3	50
180 – 185	e = 55 – 50 = 5	55
185 – 190	5	f = 55 + 5 = 60
	g = 15 + 20 + 12 + 3 + 5 + 5 = 60	

Multiple Choice Questions (MCQ)

Solution 1: (d) 26

We have:

Maximum value = 32

Minimum value = 6

We know:

Range = Maximum value − Minimum value

$= 32 - 6$

$= 26$

Solution 2: (b) 110

$$\text{class mark} = \frac{\text{upper limit} + \text{lower limit}}{2}$$

$$= \frac{120 + 100}{2}$$

$$= \frac{220}{2}$$

$$= 110$$

Solution 3: (b) 20−30

This is the continuous form of frequency distribution. Here, the upper limit of each class is excluded, while the lower limit is included. So, the number 20 is included in the class interval 20−30.

Solution 4: (b) 17.5−22.5

We are given frequency distribution 15, 20, 25, 30,...

Class size = 20 − 15 = 5

Class marks = 20

Now,

Lower limit $= \left(20 - \frac{5}{2}\right) = \frac{40-5}{2} = \frac{35}{2} = 17.5$

Upper limit $= \left(20 + \frac{5}{2}\right) = \frac{40+5}{2} = \frac{45}{2} = 22.5$

Thus, the required class is 17.5−22.5.

Solution 5: (b) 7

Given:

Mid value of the class = 10

Width of each class = 6

Now,

Let the lower limit be x.

We know:

Upper limit = Lower limit + Class size

$= x + 6$

Also,

Mid value = 10

$\Rightarrow \frac{x+6+x}{2} = 10$

$\Rightarrow 2x + 6 = 20$

$\Rightarrow 2x = 14$

$\Rightarrow x = 7$

Thus, the lower limit is 7.

Solution 6: (a) 37–47

Let the lower limit be x.

Here,

Class size = 10

$\therefore$ Upper limit = Class size + Lower limit

$= (x + 10)$

Mid value of the class interval = 42

$\therefore \frac{(x+10)+x}{2} = 42$

$\Rightarrow 2x + 10 = 84$

$\Rightarrow 2x = 74$

$\Rightarrow x = 37$

Thus, we have:

Lower limit = 37

Upper limit = 37+10 = 47

Solution 7: (a) $2m - u$

Given:

Mid value = m

Upper limit = u

We know:

$$\frac{\text{Lower limit} + \text{Upper limit}}{2} = \text{Mid value}$$

$$\Rightarrow \frac{\text{Lower limit } + \text{u}}{2} = \text{m}$$

$\Rightarrow$ Lower limit + u = 2m

$\Rightarrow$ Lower limit = 2m − u

Solution 8: (c) 35

We have:

Class width = 5

Lower class limit of the lowest class = 10

Now,

Upper class limit of the highest class $= 10 + 5 \times\times 5 = 35$

Solution 9: (c) $2m - L$

$$\text{Mid value} = \frac{\text{Lower limit} + \text{Upper limit}}{2}$$

$$\Rightarrow \mathrm{m} = \frac{\mathrm{L} + \mathrm{U}}{2}$$

$$\Rightarrow 2\mathrm{m} = \mathrm{L} + \mathrm{U}$$

$$\Rightarrow \mathrm{U} = 2\mathrm{m} - \mathrm{L}$$

Upper class boundary of the class $= 2\mathrm{m} - \mathrm{L}$